bimm143_lab5datavisR

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Background

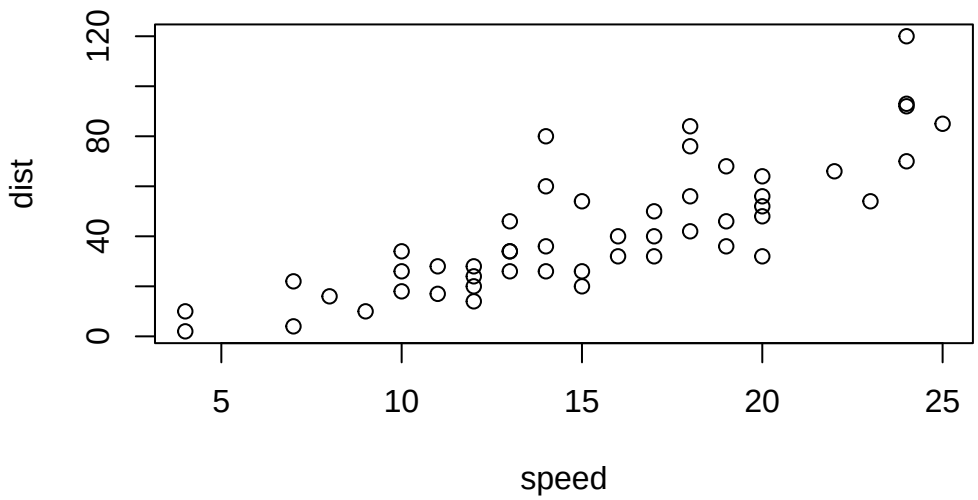
There are lots of ways to make figures in R. These include so-called "base-R" graphics. (e.g. `plot()`) and tones of add-on packages like **ggplot2**.

For example, here we make the same plot with both:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```

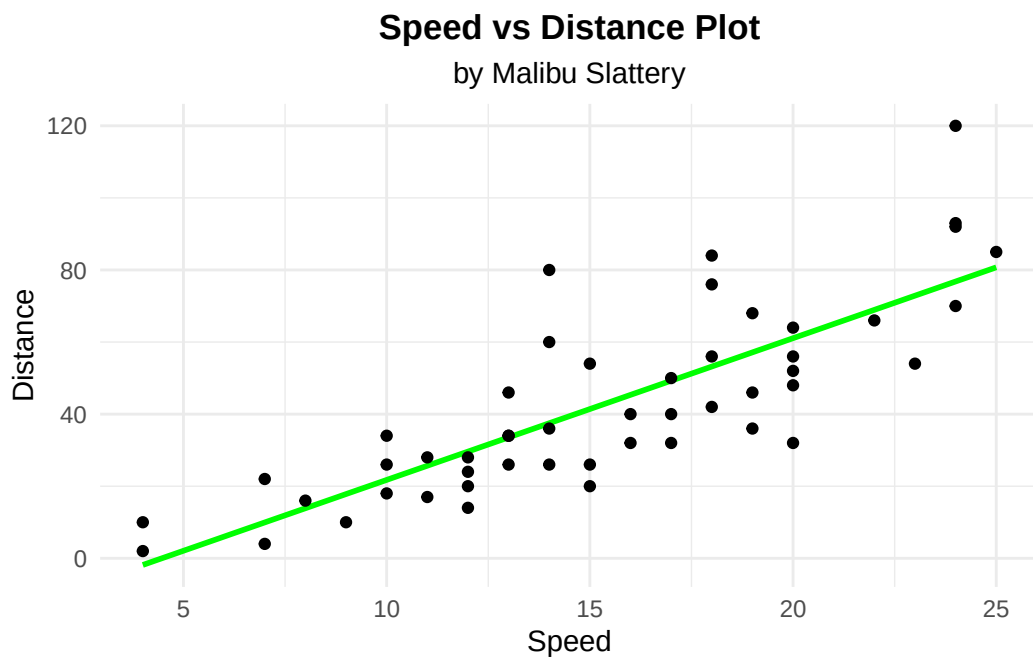



```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
p<- ggplot(cars) + aes(speed, dist) + geom_smooth(col="green", method = "lm", se = FALSE) + 
p
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene expression plot

```
library(ggplot2)
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
nrow(genes)
```

```
[1] 5196
```

```r
ncol(genes)
```

```
[1] 4
```

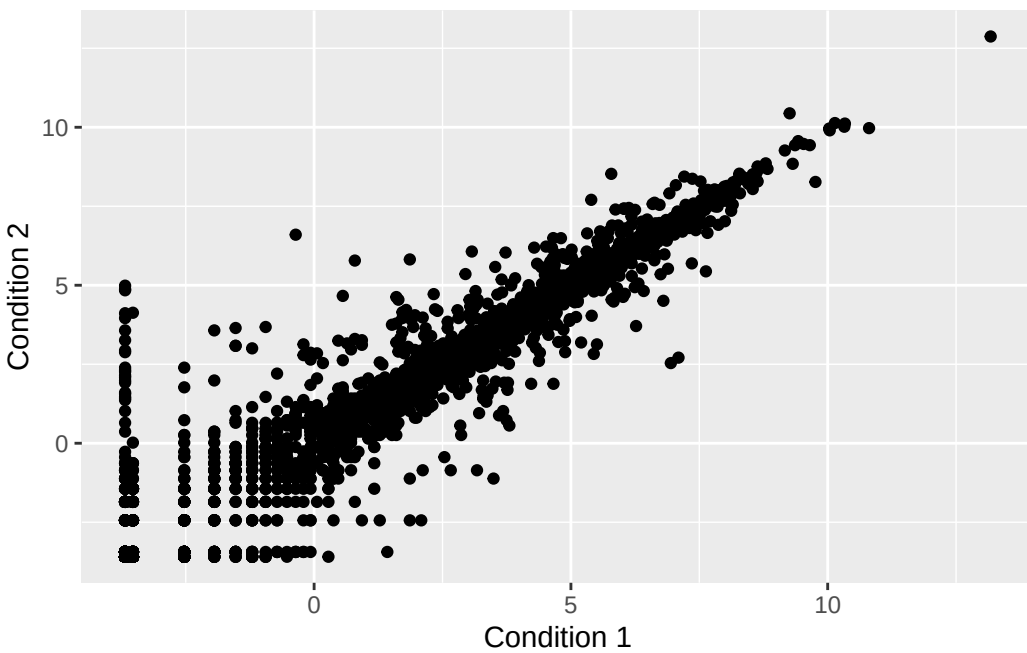
```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

```r
round( table(genes$State)/nrow(genes) * 100, 2 )
```

```
      down unchanging         up 
      1.39      96.17       2.44 
```

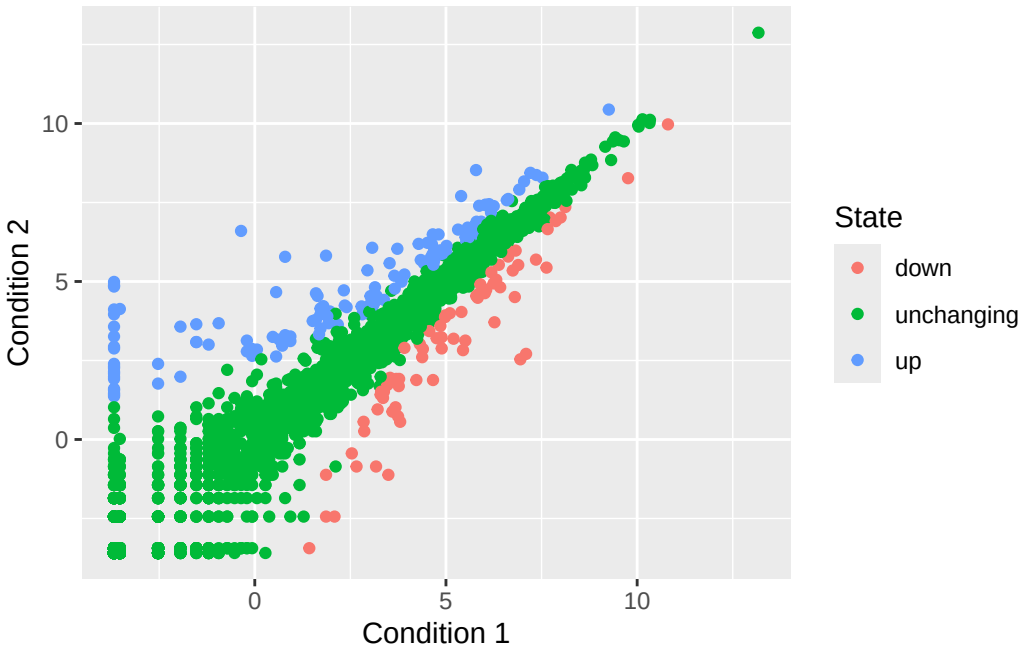
```r
ggplot(genes) +
  aes(Condition1, Condition2) +
  geom_point() + xlab("Condition 1") + ylab("Condition 2")
```



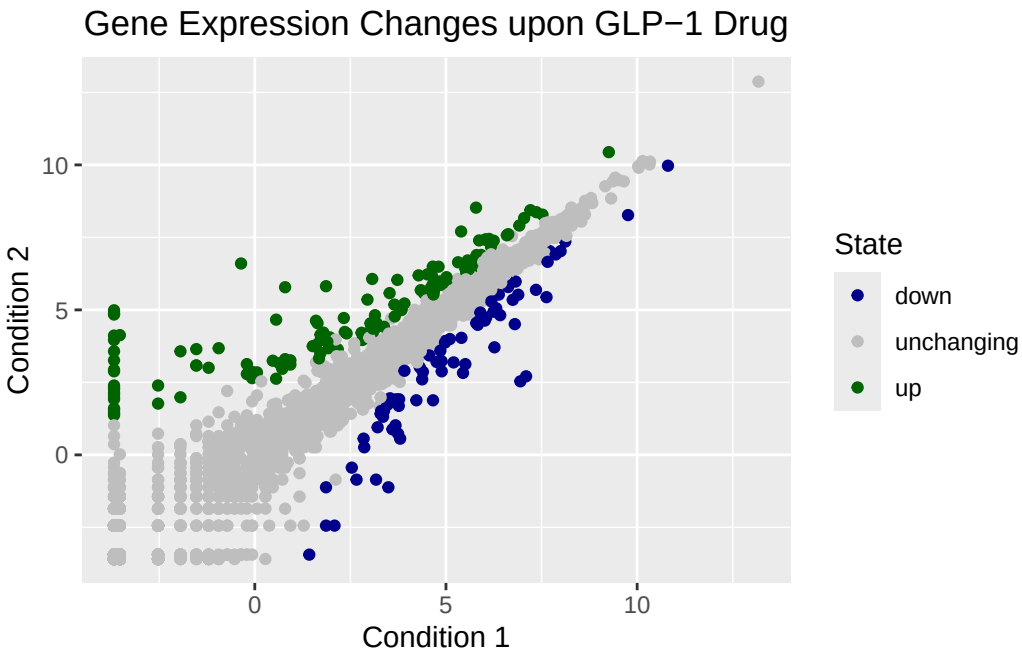
```
#genes$State
```

Condition 2: Let's color by `State`, so we can see the "up" and "down" significant genes compared to all the "unchanging" genes..

```
ggplot(genes) +
  aes(Condition1, Condition2, col = State) +
  geom_point() + xlab("Condition 1") + ylab("Condition 2")
```




```
ggplot(genes) +
  aes(Condition1, Condition2, col = State) +
  geom_point() + xlab("Condition 1") + ylab("Condition 2") + scale_color_manual(values = c("
```



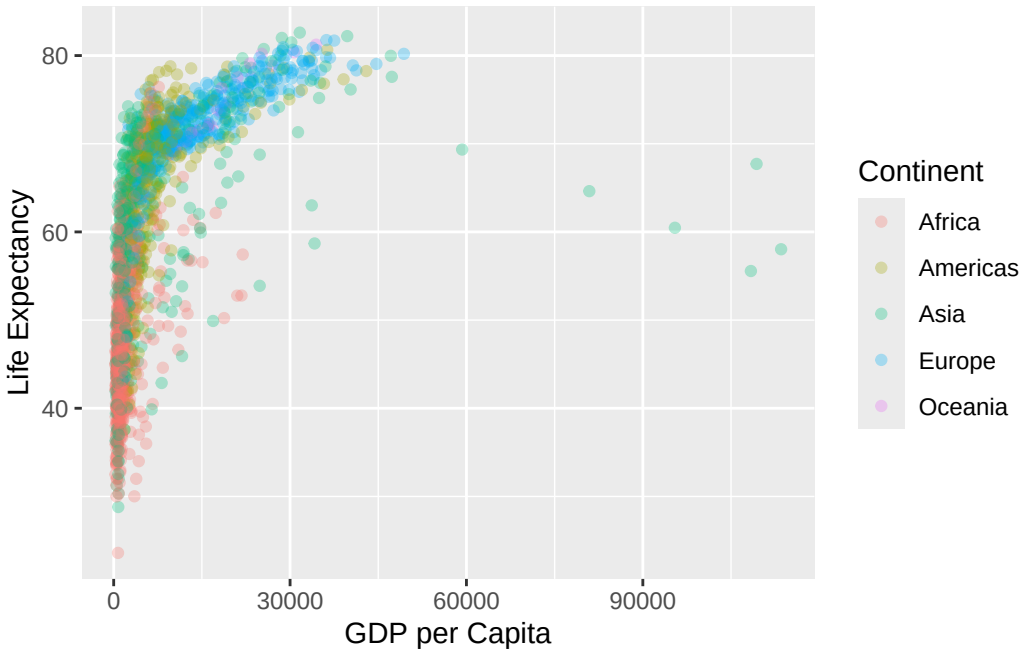
Let's look at the famous **gapminder**!

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder, 3)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
```

```
ggplot(gapminder) + aes(gdpPercap, lifeExp, col = continent) + geom_point(alpha = 0.3) + xlab
```



Let's "facet" (i.e. make a separate plot) by continent rather than the big hot mess above!

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
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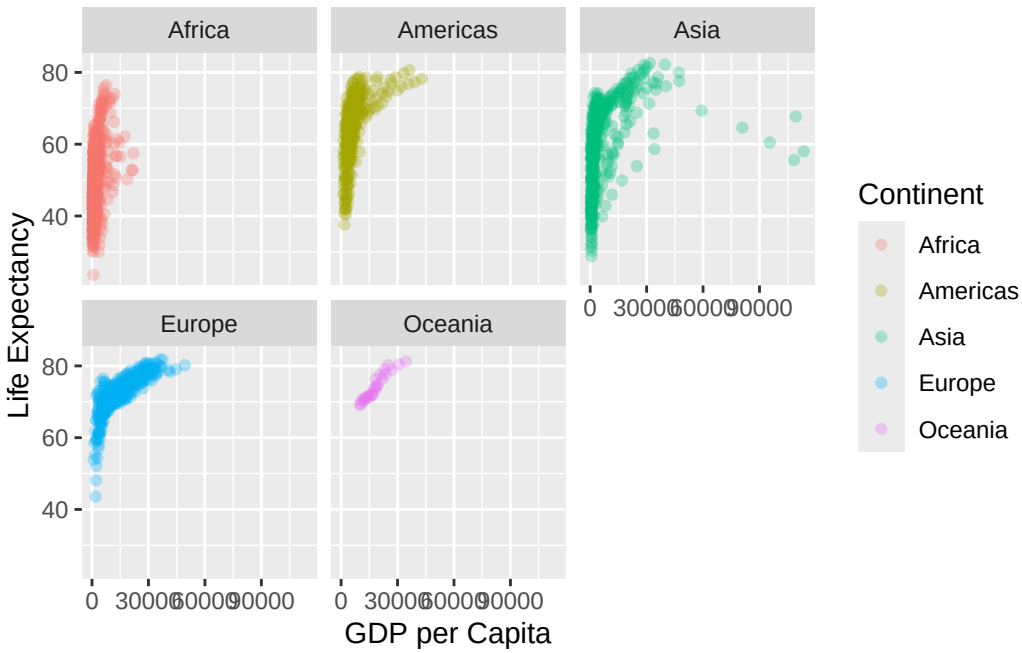
```
ggplot(gapminder) + aes(gdpPercap, lifeExp, col = continent) + geom_point(alpha = 0.3) + xlab
```



Custom Plots

How big is this gapminder dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

```
ncol(gapminder)
```

```
[1] 6
```

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
gapminder_2007<-filter(gapminder, year == 2007)
head(gapminder_2007)
```

```
      country continent year lifeExp      pop  gdpPercap
1 Afghanistan      Asia 2007  43.828 31889923   974.5803
2     Albania    Europe 2007  76.423  3600523  5937.0295
3     Algeria    Africa 2007  72.301 33333216  6223.3675
4      Angola    Africa 2007  42.731 12420476  4797.2313
5   Argentina  Americas 2007  75.320 40301927 12779.3796
6   Australia   Oceania 2007  81.235 20434176 34435.3674
```

```r
filter(gapminder, year==2007, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

```r
filter(gapminder, year==1977, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 1977   72.03 3271900  11150.98
```

```r
filter(gapminder, year==1977, country == "United States")
```

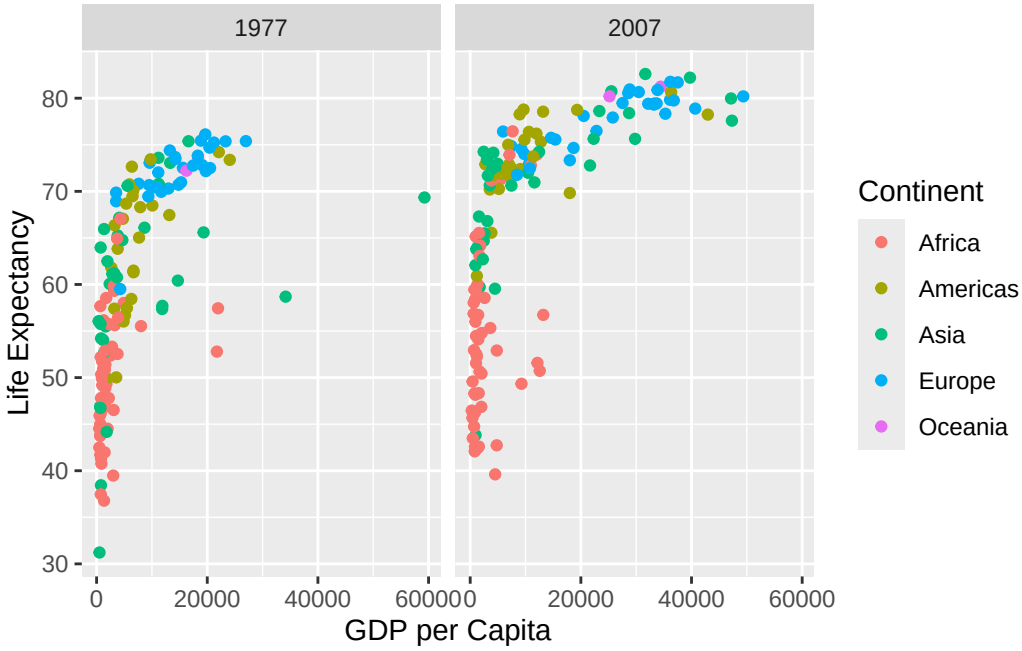
```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 1977   73.38 220239000  24072.63
```

```r
filter(gapminder, year==1977, country == "United States")$lifeExp
```

```
[1] 73.38
```

Make a plot comparing 1977 and 2007 for all countries!

```r
year_1977_2007<- filter(gapminder, year %in% c(1977, 2007))
head(year_1977_2007)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1977  38.438 14880372  786.1134
2 Afghanistan      Asia 2007  43.828 31889923  974.5803
3     Albania    Europe 1977  68.930  2509048 3533.0039
4     Albania    Europe 2007  76.423  3600523 5937.0295
5     Algeria    Africa 1977  58.014 17152804 4910.4168
6     Algeria    Africa 2007  72.301 33333216 6223.3675
```

```r
ggplot(year_1977_2007) + aes(gdpPercap, lifeExp, col = continent) + geom_point() + facet_wrap
```